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Large language models as second reviewers for medical errors in real-world internal medicine reports: a prospective comparative study of open- and closed-source models

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ABSTRACT

Objective: Preventable errors in clinical documentation and decision-making remain a major threat to patient safety, yet the role of open-source large language models (LLMs) as practical "second reviewers" in general Internal Medicine remains unclear.

Methods: We prospectively assembled 102 real-world Emergency Internal Medicine reports (de-identified) and either inserted or confirmed realistic errors across four categories (diagnostics/investigations, medication/therapy, process/communication/follow-up, other). Three LLMs (open-source Deepseek-v3-r1 and GPT-OSS-120b, and closed-source OpenAI-o3) were prompted with a uniform system instruction to (i) localize the predefined error and (ii) recommend corrections. Two blinded Internal Medicine specialists independently graded outputs for error localization (0–1) and recommendation quality (Likert 1–4); disagreements were resolved analytically, and analyses used the more conservative rater. Three human clinicians independently reviewed subsets of the same cases to provide a comparator.

Results: Using the conservative rater, correct error localization was 72.5% (74/102; 95% CI 63.2–80.3) for Deepseek-v3-r1, 79.2% (80/101; 95% CI 70.3–86.0) for o3, and 65.7% (67/102; 95% CI 56.1–74.2) for GPT-OSS-120b (Cochran's Q p = 0.033). Pairwise McNemar tests favored o3 over GPT-OSS-120b (p = 0.020; Holm-adjusted p = 0.060); other contrasts were not significant. Recommendation quality was high for all models (median 4/4), with mean ± SD scores of 3.73 ± 0.49 for Deepseek-v3-r1, 3.65 ± 0.64 for o3, and 3.51 ± 0.73 for GPT-OSS-120b. Inter-rater agreement was excellent for GPT-OSS-120b (κ = 0.94 for detection; κ_w = 0.85 for quality), substantial for Deepseek-v3-r1 (κ = 0.75; κ_w = 0.47), and lower for o3 (κ = 0.31; κ_w = 0.14). All models frequently flagged additional clinically useful issues (≥99% of reports).

Conclusion: In real-world Internal Medicine reports with realistic, expert-defined errors, state-of-the-art open-source LLMs approached the performance of a leading closed model and clearly outperformed clinicians in error detection, while providing predominantly guideline-concordant corrective recommendations. Given their advantages for privacy, customizability, and potential local deployment, open models represent credible candidates for privacy-preserving "second-reviewer" support in Internal Medicine. Prospective, workflow-embedded trials that also quantify specificity on error-free notes, alert burden, and patient outcomes are now warranted.

1. Introduction

Medical errors in clinical documentation and decision-making remain a major patient-safety challenge, spanning diagnostic omissions, medication errors, and failures in communication or follow-up [1]. In high-throughput settings such as emergency internal medicine,

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time pressure and technology-mediated documentation (e.g., templating or speech recognition) can increase the risk that clinically meaningful discrepancies persist into finalized notes [2]. Preventable harm remains common across care settings, underscoring the need for scalable, workflow-compatible safeguards that can support clinicians without adding substantial burden [3,4].

Large language models (LLMs) offer a promising approach for automated review of unstructured clinical text, with early studies showing utility for proofreading and safety checks in radiology and for medication-review tasks across specialties [5]. However, reported performance varies by model class and evaluation design, and most work has either focused on highly structured reports (e.g., radiology) or relied primarily on closed-source systems [6–12]. In parallel, open-source models are increasingly relevant for healthcare because they can, in principle, be deployed on institutional infrastructure, supporting governance and privacy requirements while allowing adaptation to local protocols. Whether contemporary open-weight models can provide clinically useful error detection in heterogeneous, free-text internal medicine documentation remains insufficiently studied.

These investigations into the performance of LLMs have yielded varied results. Comparative studies show that different models and implementation strategies, such as zero-shot prompting, few-shot learning, or fine-tuning, can produce different levels of accuracy [13]. Some analyses indicate that closed-source models may achieve higher error detection rates than open-source alternatives, though the latter offer a potential solution to data privacy concerns by allowing for local deployment [6]. In certain contexts, LLMs have demonstrated capabilities that meet or exceed human performance, particularly in terms of speed and the detection of specific error types [2]. Furthermore, integrating LLMs as a "co-pilot" tool to assist healthcare professionals has been shown to enhance performance in tasks like medication chart reviews [10]. Moreover, our previous studies have shown good efficacy for LLMs in clinical decision support, in both Ophthalmology and Internal Medicine, with models like o1 and GPT-4 reaching performance on par with human experts [14,15].

In this study, we intentionally focus on large reasoning models (LRMs)—LLMs optimized for multi-step clinical reasoning. Our primary objective is to test whether open-weight LRMs can approximate a strong closed-source LRM when used as a "second reviewer" to detect clinically relevant errors in real-world internal medicine reports. We therefore conducted a prospective comparative study using 102 de-identified emergency internal medicine notes containing expert-defined, realistic, clinically meaningful errors, and evaluated two open-weight LRMs (DeepSeek-v3-R1 and GPT-OSS-120B) against a closed-source LRM (o3). By comparing error localization and the quality of corrective recommendations under a uniform prompting and blinded clinician grading framework, we aim to inform the feasibility of privacy-preserving, locally deployable LRM-based second-review support for internal medicine documentation and decision-making.

2. Methods

2.1. Study design and patient data

This is a comparative study with a goal of evaluating and comparing state-of-the-art open source and closed source LLMs in medical error detection using real-world internal medicine patient data. The data was collected prospectively in a period from 1th of February to 10th of February 2025, at the Emergency Internal Medicine Ward of University Hospital of Split. Afterwards, the models were prompted in a retrospective fashion, ie. only after the entire dataset was collected. Patient selection criteria was that the patient report contains complete information about 1) signs and symptoms, 2) previous medical history, 3) physical examination results, 4) diagnostic tests, 5) treatment and recommendations. In this manner we have gathered a total of 102 patient reports. All patient data was de-identified before LLM prompting. All direct identifiers such as names, social security numbers, time of admittance and contact details were removed. The study protocol was reviewed and approved by the Ethics Board of the University Hospital of Split (approval number: 2181–147/01-06/LJ.Z.-24-02).

Two expert internal medicine physicians (with 6 and 10 years of work experience) have either inserted a medical error into the patient report, or have located an already present error in the report. Before this step, they were explicitly instructed to introduce only errors they could realistically encounter in routine clinical practice and to avoid contrived or obviously unrealistic scenarios. Errors therefore spanned from a missing recommendation or diagnostic test, to wrong medication, wrong dosage or similar, with a goal that such errors/mistakes reflect what is expected in a real-world clinical practice. For each candidate error, the two internists jointly reviewed whether leaving it uncorrected could plausibly change investigations, treatment, or short-term patient safety; only such "clinically relevant" errors were retained, whereas purely typographical, formatting or stylistic issues were excluded. This constitutes a simple harm framework that dichotomised potential impact into clinically relevant vs. trivial. To check how human clinicians compare against the models, we have also tested how many inserted mistakes can three human clinicians locate in the mistakes dataset.Of the 102 predefined errors, 99 (97.1%) were deliberately inserted into otherwise accurate reports by the two internists, and 3 (2.9%) corresponded to clinically relevant errors that were already present in the original documentation and were simply flagged. We did not use a formal numeric severity scale; instead, both internists restricted errors to those judged a priori to be clinically relevant (i.e., if left uncorrected, they could plausibly alter downstream investigations, therapy or short-term patient safety). Trivial typographical or formatting issues were explicitly excluded. Errors were categorised into four types (diagnostics/investigations, medication/therapy, process/communication/follow-up, and other), as detailed in the Results. The same two internists later graded the LLM outputs but did so as independent raters, blinded to model identity and to each other's assessments, using a predefined grading rubric focused on (1) correct identification of the predefined error and (2) adequacy and safety of the proposed correction; they did not redefine which errors counted as ground truth.

The study aims to elucidate whether SOTA open source and closed source LLMs can reliably detect medical errors in real-world Internal Medicine patient reports. Hence, the goal was to 1) given the inserted or located medical error; 2) prompt the LLM to identify the error, and to provide the necessary recommendations on how to best resolve the error (Fig. 1.).

2.2. LLM error detection in patient reports

Model selection was hypothesis-driven: we chose o3 as a representative closed-source reasoning model and DeepSeek-v3-R1 and GPT-OSS-120B as contemporary open-weight reasoning models to directly test open-vs-closed performance under a controlled prompting and grading protocol.

We prompted three different LLMs/LRMs: (1) DeepSeek-v3-R1 (API identifier: "deepseek-reasoner", DeepSeek AI; open-source LRM), (2) the **o3** model (API identifier: "o3″; closed-source LRM from OpenAI), and (3) GPT-OSS-120B (API identifier: "gpt-oss-120b"; open-source LRM from OpenAI, accessed via the Groq API). DeepSeek-v3-R1 was accessed through the DeepSeek API, the o3 model via the OpenAI API, and GPT-OSS-120B via the Groq API, which hosts open-source models. All models were accessed between 15 February and 10 March 2025, we used the public API versions available at that time.

For each patient report, we used a single-turn prompt flow. The core instruction (identical for all models) asked the model to act as an expert Internal Medicine doctor with $>$ 20 years of clinical experience, to analyse the full patient report for diagnostic omissions (missing investigations or procedures) and treatment/recommendation mistakes, and to return each detected error in a structured format:



Fig. 1. Methodology workflow. De-identified Internal Medicine reports (n = 102) were collected prospectively and then subjected to error insertion/identification by two experienced internists to mirror real-world documentation and clinical decision errors (diagnostic/investigations, medication/therapy, process/communication/follow-up, other). Each report was analyzed with a uniform system prompt for error detection by three large reasoning models: GPT-OSS-120b (open-source), o3 (closed-source), and Deepseek-v3-R1 (open-source). Two additional internists, blinded to model identity, independently graded outputs for (i) correct localization of the predefined error (binary) and (ii) overall recommendation quality on a 1–4 Likert scale; disagreements were resolved analytically, and inter-rater agreement was quantified prior to primary analyses. This pipeline operationalizes a head-to-head comparison of open- vs closed-source models for clinically meaningful error detection and resolution in real-world internal medicine documentation.

– "**Spotted error**: [description of the specific error]".

– "**Recommendation**: [professional recommendation on how to correct the error]".

If multiple errors were found, the model was instructed to list each separately; if no errors were found, it was instructed to output "No medical errors identified in this report." Finally, each model was asked to explicitly identify the single major mistake (if any) and provide a corresponding recommendation, because our primary endpoint was detection of the predefined major error. The full verbatim prompt text is provided in the Supplementary Materials, but all essential instructions are summarised here. No tool-calling, retrieval, or function-calling features were enabled for any model; all outputs were free-text only.

Decoding parameters were kept intentionally simple and fixed across the dataset. For GPT-OSS-120B we used a temperature of 0.3, with top_p left at the API default and without specifying max_tokens to avoid unwanted truncation of longer responses. For DeepSeek-v3-R1 and o3 we again relied on the API defaults for sampling (temperature setting is not supported, top_p parameter was set to default, and max_tokens was left unspecified). All calls were made using Python 3.11 with the vendor-specific libraries (deepseek, openai, groq) and requests.

2.3. LLM error detection and recommendation quality evaluation

Two independent expert internal medicine physicians (6 and 10 years of experience) evaluated all LLM outputs. First, they scored whether the model correctly localized the predefined error in each report (binary 0/1). Second, they noted whether the model identified any additional clinically useful issues or recommendations beyond the primary error (binary 0/1). Finally, they rated overall detection/recommendation quality on a 1–4 Likert scale: 1 = missed error and/or clearly unsafe or incorrect key advice; 2 = partially addressed problem with incomplete, non–actionable, or low-risk inaccuracy; 3 = correct error localization with a generally appropriate, safe but slightly incomplete or suboptimally prioritized recommendation; 4 = precise localization with a complete, guideline-concordant, actionable plan including what to change and relevant safety checks.

Both graders worked independently and were blinded to model identity. Inter-rater agreement was quantified using Cohen's κ for binary endpoints and quadratic weighted κ for the 1–4 grades, showing fair to excellent agreement across models (κ range 0.31–0.94; κ_w 0.14–0.85). To limit the impact of grader leniency, primary analyses used the more conservative rater, defined by lower overall detection positivity and lower mean Likert score (grader 2). This predefined choice was intended to provide a lower-bound estimate of model performance (i.e., avoiding spuriously high results due to lenient scoring) while maintaining straightforward interpretability of the ordinal 1–4 Likert rubric without introducing additional aggregation assumptions (e.g., averaging or consensus-only scoring).

We additionally analyzed a separate clinician benchmark dataset to compare human and LLM performance. Three independent physicians each reviewed approximately one third of the cases; the endpoint was correct identification of the inserted mistake (0/1), summarized as the percentage of correct detections with Wilson 95% confidence intervals.

2.4. Statistical analysis

Analyses were conducted at the level of individual clinical cases per model. The binary endpoint was correct localization of the medical error (1/0), and the ordinal endpoint was the overall detection/recommendation grade (1–4 Likert scale). Because all models rated the same set of cases, inferential procedures used paired/repeated-measures designs.

For the binary endpoint, model performance was summarized as the percentage of correctly located mistakes with Wilson 95% confidence intervals. Global differences among models were tested with Cochran's Q on complete cases; when significant, pairwise comparisons used two-sided exact McNemar tests on discordant pairs. For each comparison we reported Cohen's g and an odds ratio on discordant cells with Haldane–Anscombe correction, with normal-approximation 95% confidence intervals.

For the ordinal 1–4 grades, we summarized mean, standard deviation, and median. Global differences were assessed with Friedman's test on complete cases, followed by two-sided Wilcoxon signed-rank tests for pairwise post hoc comparisons. Effect sizes included rank-biserial correlation for pairwise tests and Kendall's W as an omnibus measure (also interpreted alongside Cochran's Q for repeated-measures binary outcomes). All tests were two-sided with $\alpha = 0.05$, and p-values for families of pairwise comparisons (binary and ordinal endpoints separately) were adjusted using the Holm–Bonferroni procedure. Statistical analyses were implemented programmatically in Python using standard numerical and scientific libraries.

3. Results

3.1. Mistakes dataset overview

Mistakes were inserted or found in a total of 102 patient reports. Furthermore, mistakes were grouped into five categories: Diagnostic/Investigations (31/102; 30.4%), Medication/Therapy (30/102; 29.4%), Process/Communication/Follow-up (15/102; 14.7%), and Other (26/102; 25.4%). Most identifiable errors involved inappropriate removal or modification of investigations (e.g., blood tests, imaging) and medications (dose/agent changes or cessation). A smaller proportion reflected process-level issues (e.g., omitted consults or patient advice) and documentation inconsistencies (e.g., altered scores or missing conclusions). Concerning the actual diagnosis distribution in the patient reports dataset, the primary diagnoses, based on the first ICD-10 code in each report, were cardiovascular in 23/102 (22.5%), respiratory in 9/102 (8.8%), gastrointestinal in 13/102 (12.7%), symptom-based presentations in 26/102 (25.5%), and observation/encounter codes in 8/102 (7.8%), with the remaining 23/102 (22.5%) classified as other

internal medicine diagnoses (infectious, hematologic, endocrine/metabolic, musculoskeletal, dermatologic, renal and miscellaneous codes) (Table 1.).

The average patient age was 61.2 years, with the youngest patient being 19 years old, and the oldest patient 91 years old. Average word count per report was 608, with the range of 241–1 281 words.

3.2. Primary Outcome (% of detected mistakes)

Using the conservative rater's assessments, model-level performance for correct localization of mistakes diverged meaningfully. The proportion correct was 72.5% (74/102; 95% Wilson CI 63.2–80.3) for Deepseek-v3-R1, 79.2% (80/101; 95% CI 70.3–86.0) for o3, and 65.7% (67/102; 95% CI 56.1–74.2) for GPT-OSS-120b (Fig. 2. A). An omnibus Cochran's Q test across models indicated heterogeneity in detection performance ($Q(2) = 6.84$, $n = 101$, $p = 0.033$), corresponding to a small repeated-measures effect by Kendall's W ($W = 0.034$). In pairwise McNemar exact tests, the strongest signal favored o3 over GPT-OSS-120b (discordant pairs $n01 = 9$ and $n10 = 23$; $p = 0.020$), which was borderline after Holm adjustment ($p_Holm = 0.060$); the effect was consistent in direction, with a discordant-pair odds ratio of 0.39 (95% CI 0.18–0.85) and Cohen's g of − 0.44. Comparisons between Deepseek-v3-R1 and o3 ($p = 0.230$; $p_Holm = 0.459$; OR = 1.78, 95% CI 0.79–4.02; $g = 0.28$) and between Deepseek-v3-R1 and GPT-OSS-120b ($p = 0.265$; $p_Holm = 0.459$; OR = 0.61, 95% CI 0.29–1.29; $g = -0.24$) did not reach statistical significance.

Three clinicians each reviewed one-third of the dataset (34 reports per clinician). Mistake detection accuracy varied across clinicians: 38.2% (13/34), 55.9% (19/34), and 50.0% (17/34), yielding an overall human detection rate of 48.0% (49/102, Wilson 95% CI: 38–54). Moreover, stratification by predefined error category showed that the largest human–LLM gap occurred for diagnostics/investigation-related errors: humans detected 36.6% (15/41) vs o3 80.5% (33/41; $p < 0.001$) and DeepSeek-R1 70.7% (29/41; $p = 0.009$). For medication/therapy errors, humans detected 51.2% (22/43) vs o3 79.1% (34/43; $p = 0.004$). Process/follow-up and "Other" strata were smaller, leading to wider confidence intervals and limited power for paired comparisons.

All three LLMs achieved statistically significant better error detection than human clinicians ($p < 0.001$, pairwise McNemar exact tests). When summarized by rater using the three-way partition, point estimates and confidence intervals were broadly comparable across the three groups, consistent with similar performance across physicians in this task.

3.3. Overall report grading (Likert 1–4)

For the ordinal detection/recommendation grade, summary statistics based on the conservative rater showed numerically higher scores for Deepseek-v3-R1 (mean 3.73 ± 0.49; median 4.0; mean 95% CI 3.63–3.82), intermediate scores for o3 (mean 3.65 ± 0.64; median 4.0; mean 95% CI 3.53–3.78), and lower scores for GPT-OSS-120b (mean 3.51 ± 0.73; median 4.0; mean 95% CI 3.37–3.65) (Fig. 2. B). The Friedman test did not show a significant omnibus difference among models ($\chi^2(2) = 5.01$, $n = 101$, $p = 0.082$), and the corresponding Kendall's W indicated a small effect ($W = 0.025$). Nonetheless, post-hoc Wilcoxon signed-rank comparisons identified a difference between Deepseek-v3-R1 and GPT-OSS-120b that remained significant after Holm correction ($p = 0.009$; $p_Holm = 0.027$), with a moderate effect size by rank-biserial correlation ($r \approx 0.32$). Differences between Deepseek-reasoner and O3 ($p = 0.388$; $p_Holm = 0.388$; $r = 0.12$) and between o3 and GPT-OSS-120b ($p = 0.141$; $p_Holm = 0.283$; $r = 0.20$) were not statistically significant.

Table 1
Distribution of primary diagnosis groups (ICD-10) in the 102 Emergency Internal Medicine cases.

Diagnosis group	ICD-10 chapters included	n	% of cases
Cardiovascular	I – Diseases of the circulatory system	23	22.5%
Respiratory	J – Diseases of the respiratory system	9	8.8%
Gastrointestinal	K – Diseases of the digestive system	13	12.7%
Symptom-based presentations	R – Symptoms, signs and abnormal clinical findings	26	25.5%
Observation/encounter	Z – Factors influencing health status and contact	8	7.8%
Other internal medicine	A, C, D, E, L, M, N, U; unspecified diagnoses	23	22.5%

3.4. Performance per mistake category and additional mistakes detection

Concerning detection of other mistakes (other then primary) and providing other recommendations the graders deemed clinically useful, Deepseek-v3-R1 scored 100.0% (95% CI 96.4–100.0; 102/102), o3 scored 99.1% (95% CI 94.7–99.8; 101/102), and GPT-OSS-120b scored 99.0% (95% CI 94.7–99.8; 101/102). These consistently high rates indicate that models frequently flagged additional minor issues (often documentation-related) that clinicians judged relevant.Across all notes, the median number of reported issues per report was identical across models (median = 7). However, dispersion differed: GPT-OSS-120B showed the tightest distribution (IQR = 2), indicating the most consistent number of alerts per note, whereas DeepSeek-R1 showed the greatest variability (IQR = 3) with occasional outliers (up to 17 issues in a single report). The o3 model had the highest upper quartile (Q3 = 9), suggesting a tendency to flag slightly more issues in the upper tail compared with the other models. Overall, models exhibited similar central tendency in alert counts but differed in variance, implying different trade-offs between consistency and potential alert load.

When stratified by error category, diagnostic/investigation-related omissions were the most difficult class (o3 27/36, 75.0%; DeepSeek-R1 24/36, 66.7%; GPT-OSS-120B 24/36, 66.7%), whereas medication/therapy errors were detected more consistently (34/42, 81.0%; 31/42, 73.8%; 32/42, 76.2%). Process/follow-up errors showed the largest variability between models (DeepSeek-R1 13/14, 92.9% vs GPT-OSS-120B 8/14, 57.1%), although this stratum was smaller (n = 14) (Table 2.). Stratification by diagnosis group showed broadly similar performance across cardiovascular, respiratory, gastrointestinal, symptom-based and observation/encounter presentations, with smaller diagnosis strata limiting power for between-model comparisons. Overall between-model differences were not statistically significant using paired nonparametric tests (Cochran's Q for detection and Friedman tests for ordinal grades; $p > 0.05$).

Taken together, these findings indicate that, under the conservative grading scheme, models differed in their ability to localize mistakes, with the clearest advantage observed for o3 relative to GPT-OSS-120b in the pairwise framework, albeit with borderline significance after multiplicity control. Grading patterns favored Deepseek-v3-R1 over GPT-OSS-120b with a moderate effect, while contrasts involving o3 did not meet conventional thresholds for statistical significance.

4. Discussion

This prospective comparative study suggests that large language models can function as "second reviewers" for real-world internal medicine documentation, identifying clinically meaningful errors and proposing largely appropriate corrective actions. Across 102 emergency internal medicine notes with expert-defined realistic errors, open-weight reasoning models (DeepSeek-v3-R1 and GPT-OSS-120B) approached the performance of a strong closed-source reasoning model (o3) for predefined error localization, while all three models exceeded the clinician comparator in this constrained single-pass proofreading setting. These findings support the feasibility of LLM/LRM-assisted safety checks as an adjunct to routine documentation

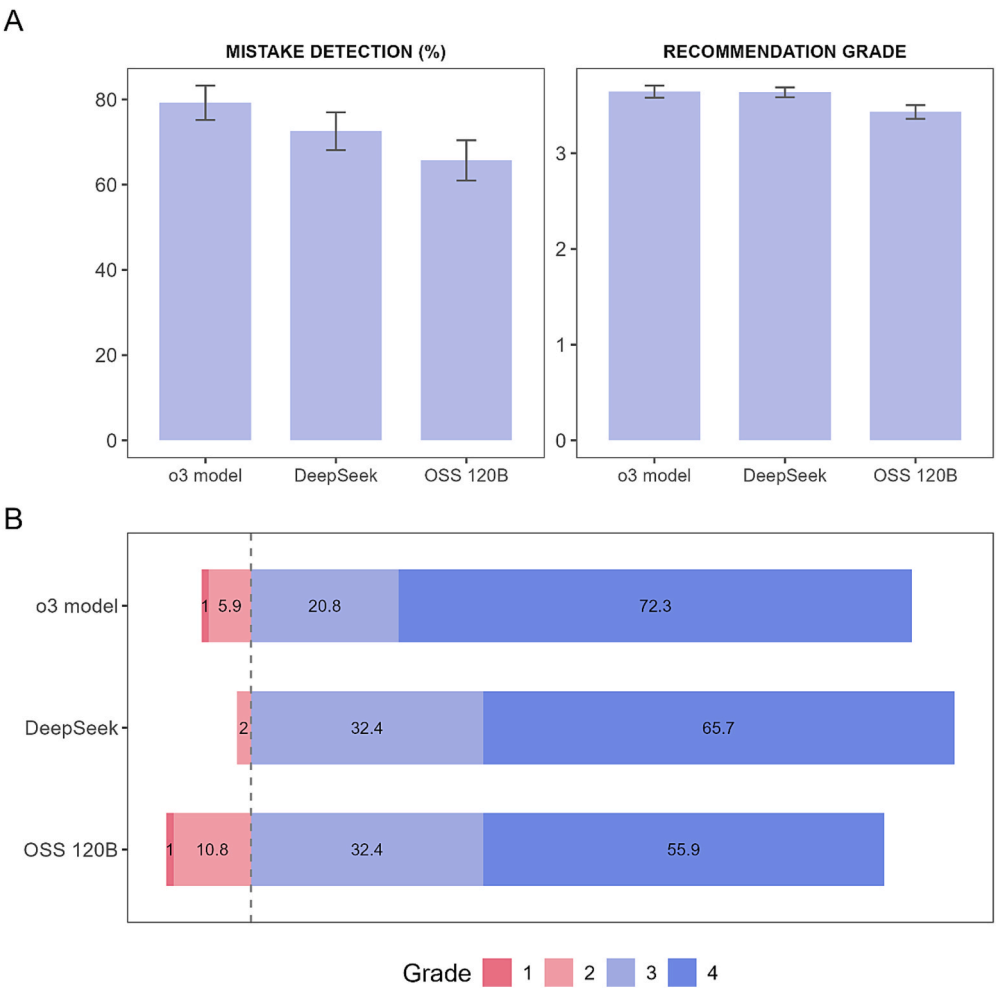


Fig. 2. Comparison of final grades for LLM error-checking efficiency. A Comparison of percentages of mistake detections and mean overall detection/recommendation grades. B Comparison of the distribution of recommendation grades over a 4-point Likert scale. The numbers within each grade denote the percentage of that grade in the total number of graded reports.

Table 2
Model performance per mistake category.

Category	n	o3 (Acc%|mean|median)	DeepSeek-R1 (Acc%|mean|median)	GPT-OSS-120B (Acc%|mean|median)	P (Accuracy)*	P (Grade)**
Diagnostics/Investigations	36	75.0%|3.61|4.0	66.7%|3.56|4.0	66.7%|3.42|4.0	0.549	0.283
Medication/Therapy	42	81.0%|3.62|4.0	73.8%|3.64|4.0	76.2%|3.67|4.0	0.607	0.934
Process/Follow-up	14	78.6%|3.71|4.0	92.9%|3.86|4.0	57.1%|3.36|4.0	0.066	0.223
Other	10	90.0%|3.90|4.0	60.0%|3.60|4.0	80.0%|3.40|3.0	0.174	0.066

*Cochran's Q test across the three models within each category.
**Friedman test across the three models within each category.

workflows, provided that outputs remain advisory and clinician-verified.

A consistent qualitative observation was that models rarely limited themselves to the single predefined error and frequently highlighted additional issues judged clinically useful by raters. This "broad net" behavior is attractive for safety, but it also raises the risk of alert fatigue if deployed without prioritization or triage. In practice, implementation will likely require interface design that separates high-severity, high-confidence concerns from lower-priority suggestions, and that supports rapid clinician confirmation rather than long narrative outputs.

Our results align with and extend prior evaluations of LLMs for clinical report review. In radiology, large-scale studies have shown that GPT-4-class systems can detect relevant reporting errors and serve as effective proofreading tools [2,6–9].

and comparative work has reported a gap between closed-source and privacy-compliant open-source models that may be narrowing as open models improve 6. Beyond radiology, LLM-assisted medication review has demonstrated measurable gains in medication safety across multiple specialties, supporting the general concept of an AI "co-pilot" that augments, rather than replaces, clinician review [10]. By focusing on heterogeneous, free-text internal medicine reports, our study contributes evidence in a less templated documentation setting where error types span diagnostic omissions, therapy issues, and process/follow-up failures.

Open-weight models are particularly relevant to healthcare because they can, in principle, support on-premise deployment and institutional governance, addressing privacy and control concerns that can limit the use of cloud-hosted proprietary systems 16,17. However, an important practical caveat is that all models in our evaluation—including the open-weight models—were accessed via cloud APIs; therefore, this study does

not directly demonstrate the operational privacy advantages of local deployment.

This study has limitations. First, the dataset size (n = 102) limits power for detecting small between-model differences. Second, we did not include error-free notes, so specificity and false-positive behavior in routine "clean" documentation remain uncertain. Third, the study was single-center and reflects local documentation style and practice patterns, which may limit generalizability. Finally, model behavior can change over time with provider updates, so results should be interpreted as a snapshot during the stated access period.

Future work should evaluate larger and more diverse datasets, include error-free notes to quantify specificity and alert burden, and test workflow-integrated deployments that measure downstream outcomes (e.g., clinician time, acceptance rates, and patient-safety endpoints). Overall, our findings support modern reasoning-capable LLMs—including open-weight models—as promising candidates for clinician-supervised second-review support in internal medicine documentation and clinical decision-making.

5. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Grammarly in for English language edits and improvements.. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Author contribution

RS designed the methodology, collected the data and was a major contributor to writing the manuscript. IV curated and de-identified the dataset, implemented the LLM prompting pipeline, and contributed to manuscript drafting and revisions. ZB performed the statistical analysis, designed the analysis plan, and led all figure generation and data visualization. MK and MV served as expert clinician raters, inserted/verified realistic errors, and interpreted clinical findings. JB provided clinical oversight, ensured methodological and ethical rigor, and critically revised the manuscript. JV conceived the study, assisted designing the methodology, coordinated the project. All authors interpreted the results, reviewed the manuscript critically for important intellectual content, and approved the final version.

CRediT authorship contribution statement

Roko Skrabic: Writing – original draft, Investigation, Data curation, Conceptualization. **Ivan Viculin:** Writing – review & editing, Validation, Formal analysis, Data curation. **Zvonimir Boban:** Writing – review & editing, Visualization, Software, Formal analysis. **Marko Kumric:** Writing – review & editing, Validation, Formal analysis, Data curation. **Marino Vilovic:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Josip Vrdoljak:** Writing – review & editing, Project administration, Methodology, Investigation, Conceptualization. **Josko Bozic:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmedinf.2026.106316.

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